

HARSH DUGAR

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Education

Sardar Patel Institute of Technology, Mumbai, India

Nov. 2022 – June 2026

Bachelor of Technology in Electronics and Telecommunication Engineering

GPA: 8.44 (Sem VI)

Technical Skills

Languages: C, C++, JavaScript, TypeScript, Python, MySQL

Development Technologies/Frameworks: HTML/CSS, React.js, Next.js, Tailwind CSS, Sanity CMS

Backend Technologies/Cloud: Node.js, Express.js, MongoDB, Cloudinary, Prisma ORM, Vercel, Render

Tools/Platforms: Git, GitHub, Linux, MicroPython, Browserless, Puppeteer, Cron

Experience

Research Intern, at Sardar Patel Institute of Technology

Jan 2025 – Present

BiLSTM Enabled Digital Stethoscope for Classifying Respiratory and Abdominal Sounds via Non-Invasive Sensing.

- **Spearheaded ML Component** of medical diagnostic research project focused on **non-invasive illness detection** through audio analysis of respiratory and abdominal sounds with **83% and 88% accuracies in Preliminary Testing**.
- Conducted extensive research and comparative analysis of **Feature Extraction** methodologies such as MFCC, Mel spectrograms, Wavelet transforms, **Data Augmentation techniques and Denoising Algorithms**.
- Currently architecting and implementing **Bi-LSTM CNN hybrid deep learning model**, combining recurrent neural networks for temporal pattern recognition with convolutional layers for feature extraction from audio spectrograms.
- Collaborating with hardware and mobile teams to integrate **ML models** into a **Full-Stack Health App**.

Projects

QueryNest: Gen-AI Driven Scraping Automation Platform [Live]

Oct 2024 - Jun 2025

Next.js, Supabase, OpenAI API, BrightData, Browserless, Puppeteer, Stripe, Clerk, Vercel

- Built a **SaaS Scraping Platform** for users to visually design, schedule, and monitor web scraping workflows via a **GenAI-powered graph-based editor**, leveraging **OpenAI** for intelligent automation and PostgreSQL with **Prisma ORM** for optimized data management.
- Integrated **Browserless** for scalable headless browser execution and **BrightData** for proxy rotation & detection evasion.
- Implemented enterprise-grade features including **Stripe Subscription Billing**, **Clerk authentication** with role-based access control, and **Cron-based job scheduling** with automatic retry mechanisms.
- Deployed on **Vercel with CI/CD pipeline**, Credit based usage and seamless **Real Time Monitoring**.

Walley: AI-Driven API Monitoring and Predictive Alert System

April 2025 - Present

- Designed and implemented **LSTM neural network for Predictive API failure detection**, providing 15+ minute advance warning with **92% accuracy in Barclays Hack-O-Hire Hackathon 2025**.
- Built **Time series forecast pipeline** via Tensorflow to analyze API response times, error rate, & resource util patterns.
- Part of comprehensive **GenAI-driven platform** featuring 5 AI components (**adaptive anomaly detection, failure prediction, cross-environment root cause analysis, smart alerts, SLA risk prediction**) designed to reduce API downtime costs and minimize alert fatigue for DevOps teams.

Acad+ : Refined Academics [Live] *MERN Stack, Tailwind CSS, Cloudinary CMS, Razorpay API*

Feb - May 2024

- Architected **Full-Stack E-Learning platform** with **Role-based access control** (Admin/Instructor/Student), custom JWT authentication, and **Razorpay integration** supporting subscriptions and purchases with automated enrollment.
- Developed robust course tools for instructors along with seamless video delivery, presented in an intuitive interface.
- Deployed on **Render** with a dynamic rating and review system enhancing course discovery and user engagement.

Positions Of Responsibility

Student Chapter - I.E.E.E. S.P.I.T — Technical Head

2023 - 2024

- **Co-Host** for eOTF Techfest involving Bombay Section, Chennai Section and Delhi Section among Others.
- Primary contributor in the **restructuring** of the **IEEE SPIT Website** during tenure reducing load time by 40%.
- Core contributor to the cross-committee organized *Bug Buster Hackathon 2024*.

Achievements and Extracurriculars

- **Finalist** at **Here Technologies Hackathon** with a Web Project on Restaurant Finder and Trip Planner App.
- **Runner Up** at the **SPIT Extempore Competition 2024** and Semi Finalist in SMP Debate.
- Extensive experience with **Linux systems**, including building and customizing installations **Arch Linux Systems**.